

MATHEMATICS

TARGET : JEE- Advanced

CAPS-11**CONTINUITY AND DIFFERENTIABILITY****SCQ (Single Correct Type) :**

- The value of $f(0)$, so that the function $f(x) = \frac{\sqrt{a^2 - ax + x^2} - \sqrt{a^2 + ax + x^2}}{\sqrt{a+x} - \sqrt{a-x}}$ ($a > 0$) becomes continuous for all x , is given by -
 (A) $a\sqrt{a}$ (B) $\sqrt{a}$ (C) $-\sqrt{a}$ (D) $-a\sqrt{a}$
- Let $f(x) = [\cos x + \sin x]$, $0 < x < 2\pi$, where $[.]$ denotes G.I.F. The number of points of discontinuity of $f(x)$ is-
 (A) 6 (B) 5 (C) 4 (D) 3
- The function $f(x) = \begin{cases} x^2 \left[\frac{1}{x^2} \right], & x \neq 0 \\ 0, & x = 0 \end{cases}$, is (where $[.]$ denotes G.I.F.)
 (A) Continuous at $x = 1$ (B) Continuous at $x = -1$
 (C) Discontinuous at $x = 0$ (D) Continuous at $x = 2$
- If $f(x) = [x^2] + \sqrt{\{x\}^2}$, where $[.]$ and $\{.\}$ denote the greatest integer and fractional part functions respectively, then-
 (A) $f(x)$ is continuous at all integral points except 0
 (B) $f(x)$ is continuous and differentiable at $x = 0$
 (C) $f(x)$ is discontinuous for all $x \in \mathbb{I} - \{1\}$
 (D) $f(x)$ is not differentiable for all $x \in \mathbb{I}$.
- If $f(x)$ is a continuous function for all real values of x satisfying $x^2 + (f(x) - 2)x + 2\sqrt{3} - 3 - \sqrt{3}f(x) = 0$, then the value of $f(\sqrt{3})$ is -
 (A) $\sqrt{3}$ (B) $1 - \sqrt{3}$ (C) $2(1 - \sqrt{3})$ (D) $2(\sqrt{3} - 1)$
- If $g(x)$ is a polynomial satisfying $g(x)g(y) = g(x) + g(y) + g(xy) - 2$ for all real x and y and $g(2) = 5$, then $\lim_{x \rightarrow 3} g(x)$ is _____.
 (A) 9 (B) 25 (C) 10 (D) none of these
- A differentiable function $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfies $f(xy) = f(x) + f(y) \quad \forall x, y \in \mathbb{R}^+$. If $f(16) = 3$, then the value of $f(2)$ is _____.
 (A) $\frac{3}{8}$ (B) $\frac{3}{4}$ (C) $\sqrt{3}$ (D) $\frac{3}{2}$



8. Let $f(x)$, $f: \mathbb{R} \rightarrow \mathbb{R}$ be a non-constant continuous function such that $f(2x) = (e^x + 1)f(x)$. the value of $f'(0)$ is _____.
- (A) $\lim_{h \rightarrow 0} \frac{f(h)}{e^h + 1}$ (B) $\lim_{h \rightarrow 0} \frac{f(h)}{e^h - 1}$ (C) $\lim_{h \rightarrow 0} \frac{f(h)}{e^h - h - 1}$ (D) $\lim_{h \rightarrow 0} \frac{f(h)}{e^{-h} - 1}$
9. If $f(x)$ be positive, continuous and differentiable on the interval (a, b) . If $\lim_{x \rightarrow a^+} f(x) = 1$ and $\lim_{x \rightarrow b^-} f(x) = 3^{\frac{1}{4}}$ also $f'(x) > (f(x))^3 + \frac{1}{f(x)}$ then ____.
- (A) $b - a > \frac{\pi}{24}$ (B) $b - a < \frac{\pi}{24}$ (C) $b - a = \frac{\pi}{12}$ (D) $b - a = \frac{\pi}{24}$
10. If $f(x) = \text{sgn}(\sin^2 x - \sin x - 1)$ has exactly four points of discontinuity for $x \in (0, n\pi)$ $n \in \mathbb{N}$ then n can be
- (A) only 4 (B) 4 or 5 (C) only 5 (D) 5 or 6

MCQ (One or more than one correct) :

11. Suppose that $f(x)$ is a differentiable invertible function with $f'(x) \neq 0$ and $h(x) = \int_1^x f(t) dx$. Given that $f(1) = f'(1) = 1$ and $g(x)$ is inverse of $f(x)$. Let $G(x) = x^2 g(x) - xh(g(x)) \quad \forall x \in \mathbb{R}$. Which of the following are correct?
- (A) $G'(1) = 2$ (B) $G'(1) = 3$ (C) $G''(1) = 2$ (D) $G''(1) = 3$
12. consider the function $f(x) = \begin{cases} \int_0^x (4 + |t-2|) dt, & x > 3 \\ ax^2 + bx & x \leq 3 \end{cases}$. If $f(x)$ is differentiable at $x = 3$, then ____.
- (A) $a + b = \frac{83}{18}$ (B) $a + b = \frac{85}{18}$ (C) $ab = \frac{7}{27}$ (D) $\frac{b}{a} = 84$
13. If $f(x) = |x^2 - 4| |x| + 3|$ then ____.
- (A) $f(x)$ is non-differentiable at 5 points (B) $f(x)$ is non-differentiable at 4 points
(C) $f(x)$ has local maxima at $x = 0$ (D) $f(x)$ has local minima at $x = -1$
14. If $f(x) = \begin{cases} a + \frac{\sin^3[x]}{x} & x > 0 \\ 3, & x = 0 \\ 2b + \left[\frac{\sin x - x}{x^3} \right] & x < 0 \end{cases}$ and $f(x)$ is continuous at $x = 0$, then
- (A) $a = 2$ (B) $a = 3$ (C) $b = 2$ (D) $b = 3$
15. If $f(x) = -1 + |x - 2|, 0 \leq x \leq 4$ and $g(x) = 2 - |x|, -1 \leq x \leq 3$ then $(f \circ g)(x)$ is ____.
- (A) Discontinuous at $x = 0$ (B) Continuous at $x = 0$
(C) Not differentiable at $x = 0$ (D) Differentiable at $x = 0$

16. Suppose f is a function that satisfies the equation $f(x+y) = f(x) + f(y) + x^2y + xy^2$ for all real numbers x and y . If $\lim_{x \rightarrow 0} \frac{f(x)}{x} = 1$, Then _____.
 (A) $f(x) > 0$ for $x > 0$ and $f(x) < 0$ for $x < 0$ (B) $f'(0) = 1$
 (C) $f''(0) = 1$ (D) $f'''(0) = 6$
17. Let $[x]$ denote the greatest integer less than or equal to x . If $f(x) = [x \sin \pi x]$, then $f(x)$ is :
 (A) continuous at $x = 0$ (B) continuous in $(-1, 0)$
 (C) differentiable at $x = 1$ (D) differentiable in $(-1, 1)$

Numerical based Questions :

18. A function $f: \mathbb{R} \rightarrow \mathbb{R}$ satisfies $\sin x \cos y (f(2x + 2y) - f(2x - 2y)) = \cos x \sin y (f(2x + 2y) + f(2x - 2y))$. If $f'(0) = \frac{1}{2}$, then the value of $4f''(x) + f(x)$ is _____
19. Let $f(x)$ be a real valued function not identically zero such that $f(x + y^3) = f(x) + [f(y)]^3 \forall x, y \in \mathbb{R}$ and $f'(0) \geq 0$, then find $f(10)$
20. Let f be a continuous function on $\mathbb{R}$ such that $f\left(\frac{1}{4x}\right) = (\sin e^x) e^{-x^2} + \frac{x^2}{x^2 + 1}$, then find the value of $f(0)$.

Matrix Match Type :

21. Match the following :

Column-1**Column-2**

(A) $f(x) = (\log_e x)(x - 1)^{\frac{1}{5}}$ at $x = 1$

(p) continuous

(B) $f(x) = [\cos 2\pi x]^4 \sqrt{\left\{ \text{som} \pi \frac{x}{2} \right\}}$, where $[.]$ and $\{.\}$ denote

(q) discontinuous

the greatest integer and the fractional part function, respectively, at $x = 1$

(C) $f(x) = \begin{cases} \cos 2x, & x \in \mathbb{Q} \\ \sin x, & x \notin \mathbb{Q} \end{cases}$ at $x = \frac{\pi}{6}$

(r) differentiable

(D) $f(x) = \min\{1, \cos x, 1 - \sin x\}$, $-\pi \leq x \leq \pi$ at $x = 0$

(s) non differentiable

(A) A-p; B-r; C-q; D-s

(B) A-r; B-p; C-s; D-r

(C) A-p, r; B-p, r; C-p, s; D-p, s

(D) A-r; B-p; C-q; D-r

Subjective Match Type :

22. Discuss the continuity of the function $f(x) = \lim_{n \rightarrow \infty} \frac{(1 + \sin x)^n + \log x}{2 + (1 + \sin x)^n}$

23. If $g(x) = \begin{cases} \frac{1 - a^x + x a^x \cdot \ln a}{x^2 a^x}, & x < 0 \\ \frac{(2a)^x - x \ln 2a - 1}{x^2}, & x > 0 \end{cases}$

(where $a > 0$), then find 'a' and $g(0)$ so that $g(x)$ is continuous at $x = 0$.

24. If $f : \mathbb{R} \rightarrow (-\pi, \pi)$ be a derivable function such that $f(x) + f(y) = f\left(\frac{x+y}{1-xy}\right)$, $xy < 1$.

If $f(1) = \frac{\pi}{2}$ and $\lim_{x \rightarrow 0} \frac{f(x)}{x} = 2$, find $f(x)$.

25. Given $f(x) = \cos^{-1}\left(\operatorname{sgn}\left(\frac{2[x]}{3x - [x]}\right)\right)$, where $\operatorname{sgn}(\cdot)$ denotes the signum function and $[\cdot]$ denotes the greatest integer function. Discuss the continuity and differentiability of $f(x)$ at $x = \pm 1$.

ANSWER KEY OF CAPS-11

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|--|--|-----------|----------|----------|
| 1. (C) | 2. (B) | 3. (B) | 4. (A) | 5. (C) |
| 6. (C) | 7. (B) | 8. (B) | 9. (B) | 10. (B) |
| 11. (AD) | 12. (BCD) | 13. (ACD) | 14. (BC) | 15. (BC) |
| 16. (AB) | 17. (ABD) | 18. (0) | 19. (10) | 20. (1) |
| 21. (C) | 22. $f(x)$ is discontinuous at integral multiples of π | | | |
| 23. $\frac{1}{\sqrt{2}}, \frac{1}{8}(\ln 2)^2$ | 24. $2\tan^{-1}x$ | | | |
| 25. f is continuous & derivable at $x = -1$ but f is neither continuous nor derivable at $x = 1$ | | | | |